

Introduction to Inferential Statistics: This are methodological to make conclusion about population from samples.

Types of Learning:

1.Supervise learning: Predicting target variables based on label data.

2.Unsupervise learning: Identifying pattern in un-label data.

Types of variables:

1.Target/Dependent variable: The variable being predicted or explained by other variable.

Notation: y.

Ex: house_price.

2.Predicted/Independent variable: The variable use to predict data or target variable.

Notation: x.

Ex: square_footage.

Key difference:

Predictor variable is the input variable i.e x.

Target variable is the output variable i.e y.

Relationship between target and predictor variable:

1.Positive relation: Predictor increases target increases.

2.Negative relation: Predictor increases target decreases.

3.Non-Correlates: Predictor has no effect on target.

Simple linear regression: Modeling relationship between one independent variable and dependent variable.

Expression: $y = \beta_0 + \beta_1 x + \epsilon$

Function:

1.lm() Linear mode:

Syntax: lm(target variable~predictor, data=dataset)

Ex:

```
data(mtcars)
```

```
model<-lm(mpg~wt, data=mtcars)
```

2.predict(): Predicting target values.

Syntax: predict(model, data.frame(predictor=value))

Ex:

```
predict(model,data.frame(wt=2.5))
```

3.plotting regression line:**Syntax:** plot()**Ex:**

```
plot (mtcars$wt, mtcars$mpg)
```

4.Add regression line:**Ex:**

```
abline(model)
```

Multiple line regression: Modeling relationship between multiple predictor and target variable.

Comparison between Single and Multiple line regression:

- 1.More than one independent variable.
- 2.Co-relation matrix helps identifying relationship.

MLR implementation in R:

lm():**Syntax:** lm(target~n no. of predictor, data=dataset)**Ex:**

```
model<-lm(mpg~wt+cyl,data=mtcars)
```

```
summary(model)
```

Co-relation matrix and F-statistic:

1.Co-relation matrix: Identifying relationship between predictor variable.

Function: cor() ->cor-relation co-efficient.

Syntax: cor(dataset[,c(n no. of predictor variable)])

2.F-statistic: Measuring model significance.

Function: summary()

Syntax: summary(model)